

3.3.2 Derivatives Worksheets

****Remember that simplifying before attempting to find the derivative is worthwhile****

1. Using Leibniz notation $\left(i.e., \frac{dy}{dx}, \frac{dP}{dt}, etc.\right)$, find the derivatives for each of the following functions, using the Product Rule when possible.

a) $y = e^x \left(\frac{1}{2}x^2 + 2x \right)$

$$\frac{dy}{dx} = e^x \left(\frac{1}{2}x^2 + 2x \right) + e^x (x + 2) = e^x \left(\frac{1}{2}x^2 + 3x + 2 \right)$$

b) $h(t) = (t - 2)(2t + 3)$

$$\frac{dh}{dt} = 1(2t + 3) + (t - 2)(2) = 2t + 3 + 2t - 4 = 4t - 1$$

c) $g(u) = \left(\frac{1}{u^2} - \frac{3}{u^4} \right) (u + 5u^3)$

$$g(u) = (u^{-2} - 3u^{-4})(u + 5u^3)$$

$$\frac{dg}{du} = (-2u^{-3} + 12u^{-5})(u + 5u^3) + (u^{-2} - 3u^{-4})(1 + 15u^2)$$

d) $y = x^{-1}e^x(x^4 - 2x + \pi)$

We have a product of 3 functions here so we use the 3-way version of the product rule.

$$\frac{dy}{dx} = -x^{-2}e^x(x^4 - 2x + \pi) + x^{-1}e^x(x^4 - 2x + \pi) + x^{-1}e^x(4x^3 - 2)$$

e) $w(r) = (1 + r + r^2)(2 - r^4)$

$$\frac{dw}{dr} = (1 + 2r)(2 - r^4) + (1 + r + r^2)(-4r^3)$$

f) $A(s) = \sqrt[3]{s}(s^2 + s + s^{-1})$

$$\frac{dA}{ds} = \frac{1}{3}s^{-2/3}(s^2 + s + s^{-1}) + \sqrt[3]{s}(2s + 1 - s^{-2})$$

2. Find the equation of the tangent line to the curve at the given value of x .

a) $y = (x+1)e^x$, $x = 0$

We calculate that $y' = 1e^x + (x+1)e^x$. We have

Point: $(0,1)$ & Slope: $m = y'(0) = e^0 + (0+1)e^0 = 2$

Therefore, the equation of the tangent line is $y - 1 = 2(x - 0) \rightarrow y = 2x + 1$.

b) $y = (x^2 - 3x + 2)(x^3 + x + 1)$, $x = 1$

We calculate that $y' = (2x - 3)(x^3 + x + 1) + (x^2 - 3x + 2)(3x^2 + 1)$. We have

Point: $(1,0)$ & Slope: $m = y'(1) = (-1)(3) + (0)(4) = -3$

Therefore, the equation of the tangent line is $y - 0 = -3(x - 1) \rightarrow y = -3x + 3$.

c) $y = e^x \sqrt{x}(x^2 - x + 2)$, $x = 1$

We calculate that $y' = e^x \sqrt{x}(x^2 - x + 2) + e^x \frac{1}{2\sqrt{x}}(x^2 - x + 2) + e^x \sqrt{x}(2x - 1)$. We have

Point: $(1, 2e)$ & Slope: $m = y'(1) = 2e + e + e = 4e$

Therefore, the equation of the tangent line is $y - 2e = 4e(x - 1) \rightarrow y = 4ex - 2e$.

3. Consider the function $y = (x^2 - 3)e^x$.

a) Determine all value of x at which the function has horizontal tangent lines.

We calculate that $y' = (2x)e^x + (x^2 - 3)e^x$.

We want all values of x for which

$$0 = (2x)e^x + (x^2 - 3)e^x \rightarrow 0 = e^x(x^2 + 2x - 3) \rightarrow 0 = e^x(x + 3)(x - 1) \rightarrow x = -3 \text{ or } x = 1.$$

So the function has horizontal tangent lines at x -values of $x = -3$ & $x = 1$.

b) Write the equations of all horizontal tangent lines found in part (a).

We have horizontal tangent lines occurring at the points $(-3, 6e^{-3})$ & $(1, -2e)$.

Since the tangent lines are horizontal they both have a slope of 0.

Therefore we have equations:

$$y - 6e^{-3} = 0(x - (-3)) \rightarrow y = 6e^{-3} \text{ at } (-3, 6e^{-3}) \text{ and}$$

$$y - (-2e) = 0(x - 1) \rightarrow y = -2e \text{ at } (1, -2e)$$

4. Determine the value of $h'(2)$, given that $f(2) = -3$, $g(2) = 4$, $f'(2) = -2$, & $g'(2) = 7$.

a) $h(x) = 5f(x) - 4g(x)$

First note that $h'(x) = 5f'(x) - 4g'(x)$.

$$\text{Therefore, } h'(2) = 5f'(2) - 4g'(2) = 5(-2) - 4(7) = -10 - 28 = -38.$$

b) $h(x) = f(x)g(x)$

First note that $h'(x) = f'(x)g(x) + f(x)g'(x)$.

$$\text{Therefore, } h'(2) = f'(2)g(2) + f(2)g'(2) = (-2)(4) - (-3)(7) = -8 + 21 = 13.$$

5. If $f(x) = \sqrt{x}g(x)e^x$ where $g(4) = 8$ and $g'(4) = 7$, find $f'(4)$.

We first calculate that $f'(x) = \frac{1}{2\sqrt{x}}g(x)e^x + \sqrt{x}g'(x)e^x + \sqrt{x}g(x)e^x$.

$$\text{Therefore, } f'(4) = \frac{1}{2\sqrt{4}}g(4)e^4 + \sqrt{4}g'(4)e^4 + \sqrt{4}g(4)e^4 = \frac{1}{4}(8)e^4 + 2(7)e^4 + 2(8)e^4 = 32e^4$$